

Maximum Empirical Likelihood Estimation

Let $y_t, t=1,2,\dots,n$, be a random sample from the population distribution $f(y, \theta)$. The parametric maximum likelihood estimation requires the knowledge of the density function, and maximizes the log likelihood function

$$\ln L = \sum_{t=1}^n \ln f(y_t; \theta)$$

The knowledge of the density function implies the knowledge about the relationship between the parameters and the density function. Consider a discrete Bernoulli distribution:

$$f(y, \theta) = \theta^{y_t}(1-\theta)^{1-y_t}, \quad y_t=0 \text{ or } 1$$

Since $p_t \equiv f(y_t, \theta) = P(y_t=1) \text{ or } P(y_t=0)$, we may consider the ML estimation as a constrained maximization

$$\max_{p_t} \sum_{t=1}^n \ln(p_t) \quad \text{s.t.} \quad p_t = \theta^{y_t}(1-\theta)^{1-y_t}$$

The distribution function contains all statistical properties of a random variable, including moments, shape, etc. Suppose that we do not have the complete information contained in the distribution function, but have only information about the moments. Can we still apply the same optimization procedure, replacing the density function with moments conditions as the constraints (i.e., *unbiased estimating function*)? For example,

$$\max_{p_t} \sum_{t=1}^n \ln(p_t) \quad \text{s.t.} \quad p_t > 0, \quad E(y_t - \theta) = 0$$

Note that we require a strict positivity for each p_t . Otherwise, the joint density will be zero. In fact this condition is already embedded in the objective function because the objective function will be negative infinity if one of the p_t is zero.

One problem with this formulation is that the objective function is not bounded, while the density function is bounded. We need a normalization constraints on p_t . We will use $\sum_{t=1}^n p_t = 1$ as the normalization. Another problem is how to compute the moment condition without knowing the distribution function. The MME computes it by the sample mean:

$$E(y_t - \theta) = 0 \quad \Rightarrow \quad \frac{1}{n} \sum_{t=1}^n (y_t - \theta) = 0$$

This is equivalent to using $(1/n)$ as the probability for each observation. The empirical likelihood estimator uses the empirical probability p_t in the place of $(1/n)$:

$$E(y_t - \theta) = 0 \quad \Rightarrow \quad \sum_{t=1}^n p_t (y_t - \theta) = 0$$

Under these two modifications we can pose the problem as

$$\max_{p_t} \sum_{t=1}^n \ln(p_t) \quad s.t. \quad p_t > 0, \quad \sum_{t=1}^n p_t = 1, \quad \sum_{t=1}^n p_t(y_t - \theta) = 0$$

Before we solve this problem, let's consider an even simpler case: we don't have even the information about the population moments. That is, we wish to find $p_t > 0$ under normalization constraint only:

$$\max_{p_t} \sum_{t=1}^n \ln(p_t) \quad s.t. \quad p_t > 0, \quad \sum_{t=1}^n p_t = 1$$

The Lagrange equation for this problem is

$$\mathcal{L} = \sum_{t=1}^n \ln(p_t) - \kappa \left(\sum_{t=1}^n p_t - 1 \right)$$

We do not need to include the inequality condition, $p_t > 0$, because it is embedded in the maximization as $\ln(0) = -\infty$. The first order conditions are

$$\frac{1}{p_t} - \kappa = 0, \quad \sum_{t=1}^n p_t = 1$$

The first condition implies $p_t = 1/\kappa$. Substituting this into the second equation, we have the solution $\hat{\kappa} = n$, and hence $\hat{p}_t = 1/n$. This indicates that the maximization of the empirical likelihood without any information about the distribution nor about moments leads to the uniform density as its estimate.

When we introduce the moment information we need to consider the following Lagrange equation

$$\mathcal{L} = \sum_{t=1}^n \ln(p_t) - \kappa \left(\sum_{t=1}^n p_t - 1 \right) + \lambda \left(\sum_{t=1}^n p_t(y_t - \theta) \right)$$

First order conditions are nonlinear in the parameters and Lagrange multipliers, and we cannot find the closed form solutions. However, it can be shown

$$\hat{\kappa} = n, \quad \hat{p}_t = \frac{1}{n - \hat{\lambda}(y_t - \theta)}$$

and $\hat{\lambda}$ is the solution to

$$\sum_{t=1}^n \frac{y_t - \theta}{n - \hat{\lambda}(y_t - \theta)} = 0$$

which can be written as a $(n-1)$ th order polynomial equation in λ . The estimate of θ can then be found from the maximization of the concentrated objective function

$$\max_{\theta} \sum_{t=1}^n \ln(\hat{p}_t)$$

 Homework #5

Consider the MEL estimator of empirical probabilities and parameter θ from a random sample $y_t, t=1,2,\dots,n$. The population moment condition is $E(y_t - \theta) = 0$. From the Lagrange equation

$$\mathcal{L} = \sum_{t=1}^n \ln(p_t) - \kappa \left(\sum_{t=1}^n p_t - 1 \right) + \lambda \left(\sum_{t=1}^n p_t (y_t - \theta) \right)$$

show that the implicit solutions are given by

$$\hat{\kappa} = n, \quad \hat{p}_t = \frac{1}{n - \hat{\lambda} (y_t - \theta)}, \quad \hat{\lambda} = \arg_{\lambda} \left(\sum_{t=1}^n \frac{y_t - \theta}{n - \lambda (y_t - \theta)} = 0 \right)$$

The example above is the case of only one empirical moment. The MELE of distribution parameter θ in a general form can be summarized as follows. We wish to maximize the empirical likelihood function

$$\max_{p_t} \ln(L) = \sum_{t=1}^n \ln(p_t)$$

subject to the strict positivity, normalization and unbiased estimating function

$$p_t > 0, \quad \sum_{t=1}^n p_t = 1, \quad E(h_t(\theta)) \equiv E[h(y_p, \theta)] = 0$$

where $h_t(\theta) \equiv h(y_p, \theta)$ is an $m \times 1$ vector for each t . Since the true distribution function is unknown, this is computed by empirical probability

$$E(h_t(\theta)) = 0 \quad \Rightarrow \quad \sum_{t=1}^n p_t h_t(\theta) = 0$$

Thus, the empirical Lagrange equation can be written as

$$\mathcal{L} = \frac{1}{n} \sum_{t=1}^n \ln(p_t) - \kappa \left(\sum_{t=1}^n p_t - 1 \right) + \lambda' \sum_{t=1}^n p_t h_t(\theta)$$

where λ is a m -dimensional vector of Lagrange multipliers.

From the first order conditions, we can derive the solution for p_t for given θ and λ :

$$\hat{p}_t(\lambda, \theta) = \frac{1}{n[1 - \lambda' h_t(\theta)]}$$

and the solution for λ for a given θ is given implicitly by

$$\sum_{t=1}^n \frac{h_t(\theta)}{1 - \lambda' h_t(\theta)} = 0$$

Substituting the solution for p_t into the empirical log-likelihood function, we can write a concentrated likelihood function as

$$\ln(L) = \sum_{t=1}^n \ln(p_t) = - \sum_{t=1}^n \ln(n[1 - \lambda' h_t(\theta)]) = - \sum_{t=1}^n \ln(n) - \sum_{t=1}^n \ln[1 - \lambda' h_t(\theta)]$$

Therefore, the original maximization problem is equivalent to

$$\min_{\theta} \sum_{t=1}^n \ln[1 - \lambda' h_t(\theta)] \quad s.t. \quad \sum_{t=1}^n \frac{h_t(\theta)}{[1 - \lambda' h_t(\theta)]} = 0$$

Note that the constraint equation that defines the solution for λ implicitly is same as the first order condition of the maximization of the following equation with respect to λ :

$$\max_{\lambda} \sum_{t=1}^n \ln[1 - \lambda' h_t(\theta)]$$

hence, we can write the problem in compact form as

$$\underset{\theta}{argmin} \max_{\lambda} \sum_{t=1}^n \ln[1 - \lambda' h_t(\theta)]$$

This MELE $\hat{\theta}$ is consistent and asymptotically normally distributed:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, Q^{-1}) \quad Q = (Eg)[E(hh')]^{-1}(Eg)'$$

where

$$h = h(Y, \theta) = \sum_{t=1}^n h(y_t, \theta)$$

$$g = \frac{dh(Y, \theta)}{d\theta} = \frac{d}{d\theta} \sum_{t=1}^n h(y_t, \theta) = \sum_{t=1}^n \frac{dh(y_t, \theta)}{d\theta} \equiv \sum_{t=1}^n g_t$$

$$est \ E(hh') = \sum_{t=1}^n \hat{p}_t[h(y_t, \hat{\theta})][h(y_t, \hat{\theta})]'$$

$$est \ E(g) = \sum_{t=1}^n \hat{p}_t g_t(y_t, \hat{\theta})$$

g_t is a $K \times m$ matrix of first order derivatives

$$g_t = \frac{dh(y_t; \theta)}{d\theta} = \left(\frac{dh_1(y_t; \theta)}{d\theta} \quad \frac{dh_2(y_t; \theta)}{d\theta} \quad \dots \quad \frac{dh_m(y_t; \theta)}{d\theta} \right)$$

Maximum Entropy Empirical Likelihood Estimator

Univariate Informational Measure

Suppose we have a set of events E_i with a probability p_i . If we receive a message that a certain event has occurred, what will be a reasonable measure of the informational content of such a message? Shannon measured it by $\ln(1/p_i) = -\ln(p_i)$. The idea is as follows. If p_i is low (e.g., $p_i = 0.01$), E_i is not expected to occur often. If E_i did occur, the news of such an occurrence is a surprise and contains a great deal of information.

This measure is not symmetric in the sense that the message of occurrence and the message of non-occurrence give different values: $\ln(1/p_i)$ and $\ln(1/(1-p_i))$. When p_i is low, the message of the occurrence of E_i is a big surprise, but the message of non-occurrence is not much of a surprise. Before we receive a message, the expected informational content of the message is called Shannon's *entropy*. When we have n events with probabilities p_i , Shannon's entropy is expressed as

$$SE(p) = \sum_{i=1}^n p_i \ln(1/p_i) = - \sum_{i=1}^n p_i \ln(p_i)$$

The maximum entropy empirical likelihood (MEEL) estimator maximizes this objective function subject to the normalization and empirical moments constraints:

$$\max_p \left(- \sum_{i=1}^n p_i \ln(p_i) \right) \quad s.t. \quad \sum_{i=1}^n p_i = 1, \quad \sum_{i=1}^n p_i h_i(\theta) = 0$$

where $h_i(\theta)$ is a $m \times 1$ vector. Setting up the Lagrange equation and solving, we get

$$\hat{p}_i(\theta, \lambda) = \frac{e^{\lambda' h_i(\theta)}}{\sum_{i=1}^n e^{\lambda' h_i(\theta)}}$$

$$\sum_{i=1}^n h_i(\theta) e^{\lambda' h_i(\theta)} = 0 \quad \Rightarrow \quad \hat{\lambda} = \lambda(\theta)$$

Thus, the MEEL estimator is obtained from

$$\underset{\theta}{argmax} \min_{\lambda} \ln \left(\sum_{i=1}^n e^{\lambda' h_i(\theta)} \right)$$

or

$$\hat{\theta}_{MEEL} = \underset{\theta}{argmax} \min_{\lambda} \left(\sum_{i=1}^n e^{\lambda' h_i(\theta)} \right)$$

Relative Entropy, Kullback-Leibler Distance (Discrepancy), Kullback-Leibler Information Criterion

In Shannon's measure of expected information content in the message is the message that states a certain event has occurred with certainty. What if the message is not certain, but states different likelihood (probability) of the events. Let q_i be the prior probabilities before the message and p_i be the posterior probabilities after the message. How can we measure the information content of such an uncertain message? One alternative is the difference in information contents between q_i and p_i , $\ln(1/q_i) - \ln(1/p_i) = \ln(p_i/q_i)$. The relative entropy of the message is the expected information contents

$$RE(p, q) = KL(p, q) = \sum_{i=1}^n p_i \ln(p_i/q_i)$$

Note that we evaluate this expectation by using the posterior probabilities p_i because that is the probability of the message. It takes a minimum value of zero when $q_i = p_i$ and can take an infinite value. This measure is called the *relative entropy of p with respect to q*, or *Kullback-Leibler distance* between p and q (*Kullback-Leibler Information Criterion*)¹. Probability p is called the *subject distribution* and q is called the *reference distribution*.

Note that, if the reference distribution is a uniform density ($q_i = 1/n$), then the KLIC becomes the negative of Shannon's entropy

$$KL(p, 1/n) = \ln(n) + \sum_{i=1}^n p_i \ln(p_i) = \ln(n) - \sum_{i=1}^n p_i \ln(1/p_i)$$

On the other hand, if the subject distribution is a uniform density ($p_i = 1/n$), then the KLIC becomes

$$KL(1/n, q) = \frac{1}{n} \sum_{i=1}^n \ln(1/nq_i) = -\frac{1}{n} \sum_{i=1}^n \ln(q_i) - \ln(n)$$

which is same as the negative empirical likelihood (EL) except for a constant term. Thus, the minimization of $KL(p, 1/n)$ is equivalent to the maximization of Shannon's entropy (MEEL), while the minimization of $KL(1/n, q)$ is equivalent to the maximization of the empirical likelihood (MEL):

$$\begin{aligned} KL(1/n, q) &= -\frac{1}{n} \sum_{i=1}^n \ln(q_i) - \ln(n) \Rightarrow \min_q KL \Leftrightarrow MEL \\ KL(p, 1/n) &= \ln(n) - \sum_{i=1}^n p_i \ln(1/p_i) \Rightarrow \min_p KL \Leftrightarrow MEEL \end{aligned}$$

For continuous distributions $f_1(x)$ and $f_2(x)$, KLIC is defined by

¹ A related concept for two random variables is the *mutual information* which is the relative entropy between the joint distribution and the product of marginal distributions (case of independent distributions). This is a measure of how much Y can tell about X.

$$KL(f_1, f_2) = \int \ln \left(\frac{f_1(x)}{f_2(x)} \right) f_1(x) dx = E_{f_1} \left\{ \ln \left(\frac{f_1(x)}{f_2(x)} \right) \right\}$$

Note that this KL information in f_1 relative to f_2 is just the expected value of the log of likelihood ratio. To be specific, suppose that the two densities are the same density with different parameter values

$$f_i(x) = f(x; \theta_i), \quad i=1,2$$

Then,

$$KL(f_1, f_2) = E_{\theta_1} \left\{ \ln \left(\frac{L(\theta_1; x)}{L(\theta_2; x)} \right) \right\}$$

Cressie-Read Distance (Discrepancy) - α -Entropy

Cressie and Reed generalized the Kullback-Leibler measure of the distance between two distributions to

$$CR(p, q, \alpha) = \frac{1}{1+\alpha} \sum_{i=1}^n p_i \frac{(p_i/q_i)^\alpha - 1}{\alpha}$$

Note that the last expression is just the Box-Cox transformation of (p_i/q_i) , which takes the value of $\ln(p_i/q_i)$ when $\alpha \rightarrow 0$. Hence, the $KL(p, q)$ is a special case¹ of the Cressie-Reed Distance (CRD) when $\alpha \rightarrow 0$:

$$CR(p, q, \alpha \rightarrow 0) = \sum_{i=1}^n p_i \ln \left(\frac{p_i}{q_i} \right) = KL(p, q)$$

which in turn implies that the MEL and MEEL are special cases of the minimization of CRD:

$$\min CR(1/n, q, \alpha \rightarrow 0) \quad \Leftrightarrow \quad MEL(q)$$

$$\min CR(p, 1/n, \alpha \rightarrow 0) \quad \Leftrightarrow \quad MEEL(p)$$

Thus, the minimization of $CR(p, 1/n, \alpha)$ is a generalization of the MEEL(p). In this case, the CRD is often expressed as

$$CR(p, 1/n, \alpha) = \frac{1}{\alpha(1+\alpha)} \sum_{i=1}^n p_i [(np_i)^\alpha - 1] = \frac{1}{n} \sum_{i=1}^n \frac{(np_i)^{1+\alpha} - 1}{\alpha(1+\alpha)}$$

Thus, the minimization of the CRD in this case becomes

$$\min_{p, \alpha} \left(\frac{1}{n} \sum_{i=1}^n \frac{(np_i)^{1+\alpha} - 1}{\alpha(1+\alpha)} \right) \quad s.t. \quad \sum_{i=1}^n p_i = 1, \quad \sum_{i=1}^n p_i h_i(\theta) = 0$$

where $h_i(\theta)$ is a $m \times 1$ vector.

¹ Since $\lim_{\alpha \rightarrow -1} \frac{r^\alpha}{\alpha(1+\alpha)} = \lim_{\alpha \rightarrow -1} \frac{r^\alpha \ln(r)}{1+2\alpha} = -\frac{1}{r} \ln(r)$, we have $CR(p, q, \alpha \rightarrow -1) = KL(q, p)$

Generalized Empirical Likelihood (GEL)

The MEL and MEEL estimators of parameter θ and probabilities p_i are summarized below:

MEL Estimation

$$\max_{p_i} \frac{1}{n} \sum_{t=1}^n \ln(p_t) \quad s.t. \quad \sum_{i=1}^n p_i = 1, \quad \sum_{t=1}^n p_t h_t(\theta) = 0$$

$$\mathcal{L} = \frac{1}{n} \sum_{t=1}^n \ln(p_t) - \kappa \left(\sum_{t=1}^n p_t - 1 \right) + \lambda' \sum_{t=1}^n p_t h_t(\theta)$$

$$p_i = \frac{1}{n[1 - \lambda' h_i(\theta)]} = \frac{[1 - \lambda' h_i(\theta)]^{-1}}{\sum_{i=1}^n [1 - \lambda' h_i(\theta)]^{-1}}$$

$$\sum_{t=1}^n \frac{h_t(\theta)}{1 - \lambda' h_t(\theta)} = 0 \quad \Leftrightarrow \quad \hat{\lambda}(\theta) = \max_{\lambda} \sum_{i=1}^n \ln[1 - \lambda' h_i(\theta)]$$

$$\hat{\theta}_{MEL} = \underset{\theta}{argmin} \max_{\lambda} \sum_{i=1}^n \ln[1 - \lambda' h_i(\theta)]$$

MEEL Estimator or Exponentially Tilted Empirical Likelihood (ETEL or ET)

$$\max_p \left(- \sum_{i=1}^n p_i \ln(p_i) \right) \quad s.t. \quad \sum_{i=1}^n p_i = 1, \quad \sum_{i=1}^n p_i h_i(\theta) = 0$$

$$\mathcal{L} = - \sum_{i=1}^n p_i \ln(p_i) - \kappa \left(\sum_{i=1}^n p_i - 1 \right) + \lambda' \left(\sum_{i=1}^n p_i h_i(\theta) \right)$$

$$p_i = \frac{e^{\lambda' h_i(\theta)}}{\sum_{i=1}^n e^{\lambda' h_i(\theta)}}$$

$$\sum_{i=1}^n h_i(\theta) e^{\lambda' h_i(\theta)} = 0 \quad \Leftrightarrow \quad \hat{\lambda} = \min_{\lambda} \left(\sum_{i=1}^n e^{\lambda' h_i(\theta)} \right)$$

$$\hat{\theta}_{MEEL} = \underset{\theta}{argmax} \min_{\lambda} \left(\sum_{i=1}^n e^{\lambda' h_i(\theta)} \right) = \underset{\theta}{argmin} \max_{\lambda} \left(\sum_{i=1}^n -e^{\lambda' h_i(\theta)} \right)$$

These two estimators can be put in a common form of Generalized Empirical Likelihood (GEL)

$$\hat{\theta}_{GEL} = \underset{\theta}{argmin} \sup_{\lambda} \sum_{i=1}^n \psi(\lambda' h_i(\theta))$$

where $\psi(v_i)$ is a function of a scalar variable $v_i \equiv \lambda' h_i(\theta)$. The GEL estimator of probabilities is defined by

$$p_i = \frac{\psi_1(v_i)}{\sum_{i=1}^n \psi_1(v_i)}$$

where ψ_1 is the first order derivative of ψ . It is clear that

$$MEL: \quad \psi(v_i) = \ln(1 - v_i), \quad MEEL: \quad \psi(v_i) = -e^{v_i}$$

Note also that the functions $\psi(v_i)$ for MEL and MEEL are concave in v_i in a open interval containing 0 and satisfy

$$\psi_1(0) = -1, \quad \psi_2(0) = -1$$

where ψ_2 is the second order derivative of ψ .

The Generalized Empirical Likelihood (GEL) considers any objective function $\psi(v_i)$ which is concave in v_i in a open interval containing 0 and satisfies the normalization conditions¹ $\psi_1(0) = -1$ and $\psi_2(0) = -1$. As shown above, the EL and EEL are special cases of GEL. Another example is the Continuous Updating Estimator (CUE) which uses a quadratic objective unction

$$CUE: \quad \psi(v) = -(1+v)^2/2$$

In the case of CR distance $CR(p, 1/n, \alpha)$, let

$$\psi(v) = -\frac{(1 + \alpha v)^{(\alpha+1)/\alpha}}{\alpha+1}$$

Then, the GEL estimator λ is the solution of

$$\hat{\lambda} = \max_{\lambda} \sum_{i=1}^n -\frac{[1 + \alpha \lambda' h_i(\theta)]^{(\alpha+1)/\alpha}}{\alpha+1} \Rightarrow \sum_{i=1}^n [1 + \alpha \lambda' h_i(\theta)]^{1/\alpha} h_i(\theta) = 0$$

which is same as the solution in the minimization of $CR(p, 1/n, \alpha)$ subject to the empirical moment

¹ Some authors impose another normalization condition $\psi(0)=0$ and specify $\psi(v)=1-e^v$ for the MEEL.

conditions. The GEL estimator of probabilities

$$p_i = \frac{\psi_1(\lambda' h_i(\theta))}{\sum_{i=1}^n \psi_1(\lambda' h_i(\theta))} = \frac{-\alpha^{-1} [1 + \alpha \lambda' h_i(\theta)]^{1/\alpha}}{-\alpha^{-1} \sum_{i=1}^n [1 + \alpha \lambda' h_i(\theta)]^{1/\alpha}} = \frac{[1 + \alpha \lambda' h_i(\theta)]^{1/\alpha}}{\sum_{i=1}^n [1 + \alpha \lambda' h_i(\theta)]^{1/\alpha}}$$

which is also same as the equation in the minimization of $CR(p, 1/n, \alpha)$ subject to the empirical moment conditions. The solution for θ is also the same because, by the envelop theorem, the solution to

$$\underset{\theta}{argmin} \sup_{\lambda} \sum_{i=1}^n \psi(\lambda' h_i(\theta))$$

is same as the first order condition

$$\frac{\partial L_{GEL}}{\partial \theta} = - \sum_{i=1}^n [1 + \alpha \lambda' h_i(\theta)]^{1/\alpha} \lambda' \frac{\partial h_i(\theta)}{\partial \theta} = 0 \quad \Rightarrow \quad \sum_{i=1}^n p_i \lambda' \frac{\partial h_i(\theta)}{\partial \theta} = 0$$

which is again same as the equation in the minimization of $CR(p, 1/n, \alpha)$ subject to the empirical moment conditions.